

“ The growth in ageing population has augmented the need for such medical equipment as biometrics, blood-pressure monitoring, blood analysis and telemedicine

—Neeraj Varma, country manager-sales, India and ANZ, Xilinx



geons using 3-D body registration systems and high-resolution imaging for cutting into invisible areas, implanted insulin pumps with a closed-loop feedback capable of automatic fine-tuning of drug delivery and thus cutting down the need for diabetics to opt for regular self-tests, and implanted stimulators and drug delivery devices that focus only on affected parts instead of oral delivery of psychiatric drugs, thus lessening to a great extent the risk of collateral damages and side effects.

With major inroads being made by IT and semiconductor technologies into the realm of healthcare, patient data storage and access has become safer and easier. Also, with patient consultation and diagnosis moving to the Internet and healthcare facilities starting to use data servers to monitor and access complete patient information over a common network, hospitals are becoming increasingly digitalised.

With digitalisation possible, hospitals can now deploy an advanced set of clinical applications like nurse call, patient monitoring and collaborative care. These help doctors and other hospital staff perform their duties with

greater efficiency and prioritise information for clinicians and their patients. Hospitals now are developing internal networks which connect all diagnostic equipment that feed complete patient information like computed tomography (CT) scans, positron emission tomography (PET) scans and X-rays over a network to data storage servers for instant access. In effect, vendors are beginning to sell complete solutions that include not only the diagnostic equipment but also the data storage servers as well as the interface software, which will eventually lead hospitals to turn ‘paperless.’

Broadly, medical devices can be divided into diagnostic and therapeutic categories. The different types of diagnostic devices are those used for recording and monitoring (ECG, EEG, blood flow and pressure), measurement and analysis (blood gas analysis, body temperature, etc) and imaging (radiation detectors and computed tomography (CT) and MRI scanners). Therapeutic devices are those which deliver physical substances (voltage, current, pressure, flow, force, ultrasound, electromagnetic radiation, heat,

etc) to the body to treat disease. These can be further categorised into devices used to treat disorders and devices to assist or control the physiological functions.

Technological trends are specifically as follows:

Imaging techniques. Imaging includes diagnostic measures for the clinical examination of human bodies. These include everything from X-rays to CT scanners, and from MRIs and ultrasound to nuclear medicines:

Ultrasound. Miniaturisation, portability, fusion imaging (merging of two different modalities like CT/MRI with ultrasound images), and faster electronic processing and data acquisition are major trends in ultrasound imaging. Software beam forming with the help of digital signal processors (DSPs) for ultrasound imaging provides flexibility and reduces hardware cost of high-end equipment.

MRI. Today’s MRI equipment offer increased patient comfort and enhanced diagnostic information. Techniques to speed up examination, such as parallel imaging technology, are implemented to speed up MRI scanning, thereby enabling shorter breath-hold times, higher resolution and shorter stay time for the patient within the magnet for any examination.

Innovations are being made for more non-invasive techniques, especially looking at possibilities to reduce the use of dyes in the procedure (which can react in some patients). High-performance magnets up to 11 Tesla are now being used. Also, today’s MRI machines are more compact and less claustrophobic.

CT. A key trend in the CT segment is shift towards combination scanners, which are primarily hybrid scanners comprising positron-emission tomographs (PETs) and CT imaging capabilities. New technologies can fuse MRI and CT images to predict the pattern of illnesses.

CT scanners started with 16 slice machines and now have gone up to 128 slice machines and even beyond. These

Market size

Healthcare is one of the largest service sectors in India. A 2009 sectoral report by National Institute of Pharmaceutical Education and Research, Ahmedabad, predicted that the medical devices and supplies market, growing at a rate of 23 per cent, would reach \$1.7 billion in 2010. High-end medical specialty electromedical equipment accounts for 11 per cent of this market demand, whereas X-ray apparatus account for 10 per cent. Imaging and electronic treatment devices make up for about Rs 13 billion and Rs 10 billion, respectively, of the medical devices market.

The main drivers for increase in the demand for medical electronics are:

1. Increase in the number of ailing people in India
2. Increase in the number of people with lifestyle diseases
3. Rising costs of healthcare in emerging markets like India
4. Rural healthcare initiatives
5. Increased adoption of miniaturisation and portability in medical devices
6. Increase in the overall consciousness of health amongst people in recent years